

EF prediction: model summary, results, and conclusions

EF prediction: what we built, results, and takeaways

Short summary of the pediatric **ejection fraction (EF)** models in `ef_prediction/`: architecture, validation numbers, and how we checked **demographics** and **embeddings**.

What the models do

- **Input:** Echo **video** (32 frames, 128×128) plus an **11-number demographic vector** (sex, age bin, BMI category).
- **Real model:** One ResNet–LSTM stream on the **real** clip; demographics are encoded and **concatenated** with the video summary before the final layers that output EF.
- **Fused model:** Same idea, but each frame uses a **learned mix** of **real** and **synthetic** video (gated fusion), then the same LSTM, demographics, and EF head.

Both models train mainly to **minimize EF error**, with an extra **contrastive-style loss** on a **video-only** side vector (not on the part that already includes demographics).

Results (validation)

EF errors are in **percentage points** (e.g. $MAE \approx \text{mean} |\text{predicted EF} - \text{true EF}|$). **Lower MAE / MSE / RMSE is better; higher R² is better** (up to 1). Re-run `evaluate_ef_real.py` / `evaluate_ef_fused.py` after retraining to refresh checkpoint metrics.

Overall (two ways we logged them)

A — Main eval JSON (single fused run vs real checkpoint)

Model	MAE	MSE	RMSE	R ²
Real	4.52	38.24	6.18	0.586
Fused (run 1, tag <code>20260419_173306</code>)	4.40	37.08	6.09	0.598

Sources: ef_prediction/eval_results/real_metrics.json,
ef_prediction/multi_run_results/fused_run_1_20260419_173306_metrics.json.

B — Same cohorts, grouped summary CSV (overall lines only)

Line	MAE	MSE	RMSE	R ²
OVERALL_REAL	4.52	38.24	6.18	0.586
OVERALL_FUSED	4.38	35.97	6.00	0.610

Source: ef_prediction/group_results/final_results_summary.csv. Numbers differ slightly from (A) for fused because the **grouped pipeline** can aggregate differently than the single-run JSON; both are valid snapshots.

Fused model — by sex

Group	MAE	MSE	RMSE	R ²	<i>n</i>
Male	3.97	26.18	5.12	0.722	6113
Female	4.24	33.12	5.76	0.764	4595

Fused model — by age bin

Group	MAE	MSE	RMSE	R ²	<i>n</i>
0–1 y	5.04	49.89	7.06	0.823	960
1–2 y	3.94	24.37	4.94	0.860	438
2–3 y	3.94	26.76	5.17	0.708	459
3–5 y	3.48	20.97	4.58	0.436	847
5–8 y	3.88	24.72	4.97	0.710	1515
8–12 y	4.02	28.55	5.34	0.734	2120
12–15 y	4.04	27.80	5.27	0.684	2322

15–18 y	4.21	29.68	5.45	0.681	2060
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Fused model — by BMI category

Group	MAE	MSE	RMSE	R ²	<i>n</i>
Underweight	4.17	30.67	5.54	0.777	5302
Normal	3.88	24.69	4.97	0.596	3643
Overweight	4.44	38.81	6.23	0.775	1029
Obese	3.87	26.23	5.12	0.686	681

Subgroup tables are from `final_results_summary.csv` (same file as OVERALL_B / fused breakdown).

How we studied demographics and plots

Embeddings (t-SNE / UMAP)

Early plots used a vector that **does not include demographics** (video-only). Coloring by sex/age/BMI looked mixed **because that vector never saw those inputs**.

Use `head_input` in the viz scripts so the 2D plot matches what the **EF head** actually receives.

Ablation

Script `ablate_demographic_effect.py`: compare **normal** vs **zeroed** vs **shuffled** demographics. If scores barely change, the model is leaning mostly on **video**; if they change a lot, demographics **matter for the prediction**.

Hypersphere (3D)

Tool `visualize_hypersphere.py`: normalize features, **PCA to 3D**, plot on a unit sphere. **EF-colored** plots tend to show **structure** (the model organizes features around the task). **Sex-colored** plots often stay **mixed** in the first 3 directions—expected when the main signal is **echo appearance**, not a separate “sex axis.”

FairHICON PDF

That paper uses **genomics** and **separate** male/female/common pathways. We did **not** copy that architecture; we only borrow ideas (e.g. hierarchical contrastive on the video side). Do not expect the same 3D “regions” unless you redesign the network and losses that way.

Main conclusions

- The pipeline predicts EF from **video**, with **demographics fused in late**; the fused variant **beats** the real model on overall MAE/MSE in the tables above (JSON and grouped summaries agree on direction).
- **Subgroup tables** show error is **not uniform** (e.g. infants 0–1 y highest MAE; several age/BMI bins lower). That is descriptive reporting, not proof that demographics *cause* those gaps.
- **Embedding color** is easy to misread: use `head_input` for “what the regressor sees,” and treat maps as **illustrations**, not proof of fairness or causality.
- To argue **whether demographics change the output**, rely on **ablation** and **subgroup errors**, not only on t-SNE/UMAP/sphere pictures.

Exports

Word, HTML, and PDF copies: `ef_prediction/documents/MODEL_RESULTS_REPORT.*`

Rebuild: `bash ef_prediction/export_report.sh`

Figures from saved checkpoints; update this file after major training or data changes.